Assignment 4 12209095 杜奕璇

1. 5.2 11.8 7.1 14.4 6.3 7.5 11.1 5.5 9.8 12.7 10.5 10.9 8.7 9.2

(a) 5.2 5.5 6.3 7.1 7.5 8.7 9.2 9.8 10.5 10.9 11.1 11.8 12.7 14.4 $n=14$

$\frac{x_{(4)}-x_{(3)}}{\frac{4}{15}-\frac{3}{15}} = \frac{\pi_{0.25}-x_{(3)}}{\frac{4}{14}-\frac{3}{15}}$ $\therefore \pi_{0.25}=6.9$

$\pi_{0.35} = \frac{3}{4}x_{(5)}+\frac{1}{4}x_{(6)} = 7.8$ $\pi_{0.75}=\frac{3}{4}x_{(11)}+\frac{1}{4}x_{(6)}=11.275$

$m = \frac{1}{2}x_{(7)}+\frac{1}{2}x_{(8)} = 9.5$

(b) i. $\pi_{0.25}$ CI: $= P_{binom}(4,14,0.25) - P_{binom}(0,14,0.25) = 0.7237$

ii. $\pi_{0.35}$ CI: $= P_{binom}(7,14,0.35) - P_{binom}(2,14,0.35) = 0.8407$

iii. m CI: $= P_{binom}(9,14,0.5) - P_{binom}(4,14,0.5) = 0.8204$

iv. $\pi_{0.75}$ CI: $= P_{binom}(12,14,0.75) - P_{binom}(7,14,0.75) = 0.8608$

(c) i. $H_0: m=8$ $H_1: m<8$

$\Leftrightarrow n=14,\ X=5,$ $H_0: P=0.5$ $H_1: P<0.5$

```
> data <- read.csv("C:\\Users\\HUAWEI\\Desktop\\sta2002\\MSFT_stock.csv")
> binom.test(5,14,alternative="less")

        Exact binomial test

data:  5 and 14
number of successes = 5, number of trials = 14, p-value = 0.212
alternative hypothesis: true probability of success is less than 0.5
95 percent confidence interval:
 0.0000000 0.6095851
sample estimates:
probability of success
             0.3571429
```

P-value $=0.212 > 0.05$

$\therefore$ Do not reject H_0

(ii) wilcox.test(data, mu=8, alternative="less", exact=FALSE)

$\rightarrow P=0.9579$ Do not reject H_0

(iii) t.test(data, mu=8, alternative="less")

$\rightarrow P=0.9526 > 0.25$

$\therefore$ do not reject H_0

2. (a)

```
> # Calculate differences from the hypothesized median
> differences <- blueberries - 9
>
> # Count the number of positive differences
> positive_diff <- sum(differences > 0)
>
> # Perform the sign test using binom.test
> result_sign_test <- binom.test(positive_diff, length(differences), p = 0.5, alternative = "greater")
>
> # Print the result
> print(result_sign_test)

        Exact binomial test

data:  positive_diff and length(differences)
number of successes = 9, number of trials = 15, p-value = 0.3036
alternative hypothesis: true probability of success is greater than 0.5
95 percent confidence interval:
 0.3595652 1.0000000
sample estimates:
probability of success
                   0.6
```

$p = 0.3036 > 0.1$

do not reject H_0

(b)

```
> # Given data
> blueberries <- c(8, 10, 7, 12, 15, 8, 10, 10, 11, 12, 9, 13, 6, 11, 6)
>
> # One-sample Wilcoxon signed-rank test
> result_wilcox <- wilcox.test(blueberries - 9, alternative = "greater", exact = FALSE)
>
> # Print the result
> print(result_wilcox)

        Wilcoxon signed rank test with continuity correction

data:  blueberries - 9
V = 71, p-value = 0.1272
```

$p = 0.1272 > 0.1$

$\therefore$ don't reject H_0

(c)

```
data:  blueberries - 9
V = 71, p-value = 0.1272
alternative hypothesis: true location is greater than 0

> # One-sample t-test
> result_ttest <- t.test(blueberries, mu = 9, alternative = "greater")
>
> # Print the result
> print(result_ttest)

        One Sample t-test

data:  blueberries
t = 1.2972, df = 14, p-value = 0.1078
alternative hypothesis: true mean is greater than 9
95 percent confidence interval:
 8.689946      Inf
sample estimates:
mean of x
 9.866667
```

$p = 0.1078 > 0.1$

$\therefore$ don't reject H_0

3. (a) $\frac{X_{(n)}}{\gamma} \leq x \Rightarrow \frac{X_{(1)}}{\gamma} < x, \frac{X_{(2)}}{\gamma} < x, \cdots$

$\therefore P\left(\frac{X_{(n)}}{\gamma} \leq x\right) = P\left(\frac{X_{(1)}}{\gamma} \leq x\right) \cdot P\left(\frac{X_{(2)}}{\gamma} \leq x\right) \cdots P\left(\frac{X_{(n)}}{\gamma} \leq x\right) = x \cdots x = x^n$

(b) $P\left(\gamma \geq \frac{X_{(n)}}{x}\right) = x^n$

let $\alpha = x^n$, $x = \alpha^{\frac{1}{n}}$

$\therefore P\left(\gamma \geq \frac{X_{(n)}}{\alpha^{\frac{1}{n}}}\right) = \alpha \Rightarrow 100(1-\alpha)\%\ CI : \left[X_{(n)}, \frac{X_{(n)}}{\alpha^{\frac{1}{n}}}\right]$

(c) $-z_{\frac{\alpha}{2}} \leq \frac{\bar{x} - \frac{\gamma}{2}}{s/\sqrt{n}} \leq z_{\frac{\alpha}{2}}$

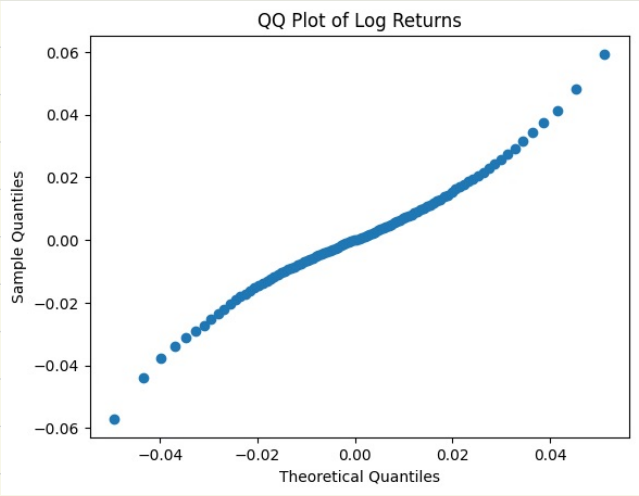
$\therefore -z_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}} \leq \bar{x} - \frac{\gamma}{2} \leq z_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$

$\therefore 2\bar{x} - \frac{2z_{\frac{\alpha}{2}} s}{\sqrt{n}} \leq \gamma \leq 2\bar{x} + \frac{2z_{\frac{\alpha}{2}} \cdot s}{\sqrt{n}}$ $\therefore 100(1-\alpha)\%\ CI$ for γ is $2\bar{x} \pm \frac{2z_{\frac{\alpha}{2}} \cdot s}{\sqrt{n}}$

4. (a)

```
Sample Mean (r): 0.0008868566305898708
Sample Variance (s^2_r): 0.00046746342193624597
```

(b)



(c) Not really

5. (a)

```
2.5663407113905564e-07 < 0.05
Reject H0: p-value = 0.000000257
```

(b) wilcox.test(log_return, mu=0, alternative = "greater", exact=FALSE)

$\Rightarrow p = 7.278 \times 10^{-6} < 0.05$

$\therefore$ reject H_0.

(c) Reject H_0: p-value = 0.000129 < 0.05